

Checking the steady frictionless solution

Substitution into the one-dimensional shallow water equations

A subscript on a field is a partial derivative. The depth is h , the bed elevation is $z(x)$, and the free-surface elevation is $\eta = h + z$. The depth-averaged velocity is u , and g is gravity. This note takes the steady frictionless flow from the companion note and substitutes it into the shallow water equations.

1. The equations

With a fixed bed and no friction, the one-dimensional system is

$$\begin{aligned} h_t + (hu)_x &= 0, \\ u_t + uu_x + g\eta_x &= 0. \end{aligned} \tag{1}$$

A steady flow does not depend on time, so $h = h(x)$, $u = u(x)$, and $z = z(x)$. Every time derivative drops out, and the system that has to be checked is

$$\begin{aligned} (hu)_x &= 0, \\ uu_x + g(h + z)_x &= 0. \end{aligned} \tag{2}$$

The check below assumes $h(x) > 0$ and that h , u , and z are differentiable, so these classical derivatives exist.

2. The candidate

The steady frictionless solution says that the discharge and Bernoulli's constant are independent of x :

$$\begin{aligned} q_0 &= hu, \\ B &= \frac{u^2}{2} + g(h + z), \end{aligned} \tag{3}$$

with q_0 and B constant. Equivalently, once a positive depth $h(x)$ and a nonzero discharge q_0 are chosen,

$$u(x) = \frac{q_0}{h(x)}, \quad z(x) = \frac{B}{g} - h(x) - \frac{q_0^2}{2gh(x)^2}. \tag{4}$$

The second formula is the bed that makes this depth a steady flow. The algebra below uses only q_0 and B being constant.

3. Mass

The depth and the velocity depend only on x , so $h_t = 0$. The discharge is the constant q_0 , so

$$(hu)_x = (q_0)_x = 0. \tag{5}$$

The mass equation is therefore

$$h_t + (hu)_x = 0 + 0 = 0. \tag{6}$$

4. Momentum

The velocity depends only on x , so $u_t = 0$. Differentiate Bernoulli's constant:

$$B_x = \left(\frac{u^2}{2} \right)_x + g(h+z)_x = uu_x + g(h+z)_x. \quad (7)$$

The left side is zero because B does not depend on x . Hence

$$u_t + uu_x + g\eta_x = 0 + uu_x + g(h+z)_x = B_x = 0. \quad (8)$$

The momentum equation holds.

5. The other direction

The same two lines run backwards, so the steady equations and the candidate describe the same smooth flows.

From $(hu)_x = 0$ and a positive depth, the product hu is a constant q_0 . From the steady momentum equation,

$$uu_x + g(h+z)_x = \left(\frac{u^2}{2} + g(h+z) \right)_x = 0, \quad (9)$$

so $u^2/2 + g(h+z)$ is a constant B . Every differentiable steady frictionless flow with $h > 0$ is of this form, and every flow of this form satisfies the shallow water equations.

6. A smooth example

The formulas fix the bed and the velocity once a depth and a discharge are chosen. Take $g = 9.81$ and $q_0 = 1$, and

$$h(x) = 1 - 0.3 \exp \left(- \left(\frac{x}{2.5} \right)^2 \right). \quad (10)$$

On the flanks the depth is one and the bed is set to zero, so $B = 1/2 + g$. The depth is at least 0.7, which lies above the critical depth $(q_0^2/g)^{1/3}$, so the Froude number stays below one. This is the smooth subcritical case covered by the substitution. The bed is a bump, the free surface dips over the crest, and $u = q_0/h$ is largest there.

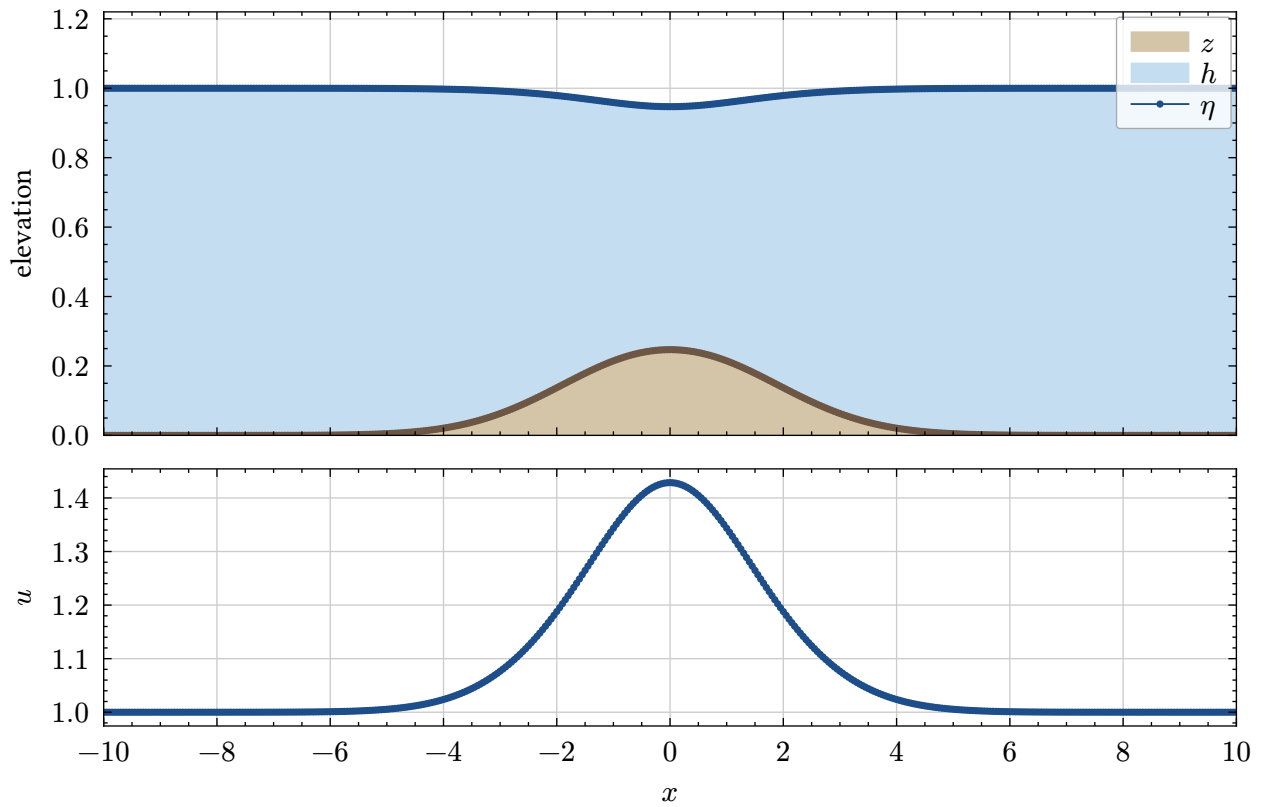


Figure 1: Bed, water, and free surface for this depth, and the velocity $u = q_0/h$.

7. Where the check stops

Across a hydraulic jump, h and u jump, so h_x and u_x are not classical derivatives and the substitution above does not apply. The jump is a weak solution when the Rankine–Hugoniot relations hold. Bernoulli's constant is not the same on the two sides of a shock. The subcritical, transcritical-without-shock, and smooth parts of a transcritical flow are covered by the substitution. The shock itself is an extra condition, not a consequence of B being constant.